



Module 7.2: Windows Architecture and Operations

- ENDPOINT SECURITY | CISCO NETWORKING ACADEMY
- PREPARED BY: KUDZAISHE MAJEZA

Hardware Abstraction Layer (HAL)

- ▶ HAL is a layer between hardware and the operating system
- ▶ Allows Windows to work with different hardware types
- ▶ Provides a standard interface for drivers and the kernel
- ▶ Improves compatibility and portability

User Mode vs Kernel Mode

- ▶ Windows operates in two main modes
- ▶ User Mode: where normal applications run
- ▶ Limited access to system resources
- ▶ Kernel Mode: core part of the OS runs here
- ▶ Has full access to hardware and memory

Why the Two Modes Matter

- ▶ Improves system stability and security
- ▶ User applications cannot directly affect hardware
- ▶ Prevents faulty apps from crashing the whole system
- ▶ Kernel handles critical tasks like memory and drivers

Windows File Systems



- ▶ File system manages how data is stored and retrieved
- ▶ Common Windows file systems: FAT32, NTFS, exFAT
- ▶ NTFS is the most widely used in modern Windows
- ▶ Supports permissions, encryption, and large files

NTFS Features



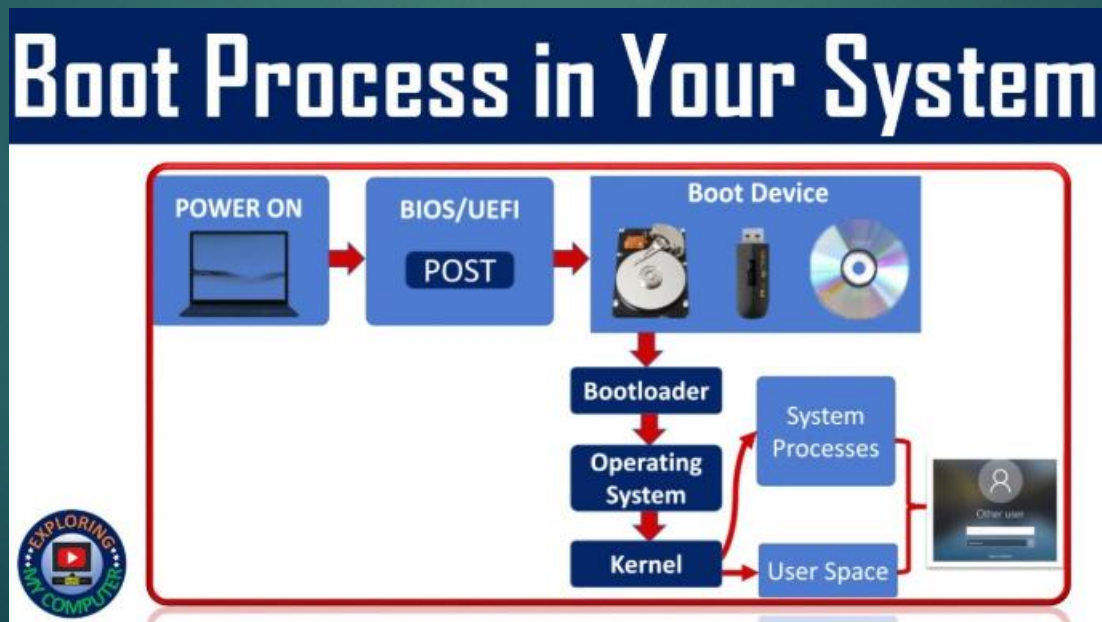
- ▶ File and folder permissions
- ▶ Disk quotas
- ▶ File compression
- ▶ Encryption using EFS
- ▶ Journaling for data recovery

Alternate Data Streams (ADS)

- ▶ Feature of NTFS file system
- ▶ Allows multiple data streams in a single file
- ▶ Can store hidden information inside files
- ▶ Sometimes used by attackers to hide malware

Windows Boot Process

- ▶ Process that starts the computer and loads Windows
- ▶ Begins when the system is powered on
- ▶ BIOS/UEFI performs hardware checks
- ▶ Bootloader loads the Windows kernel
- ▶ Kernel initializes system components



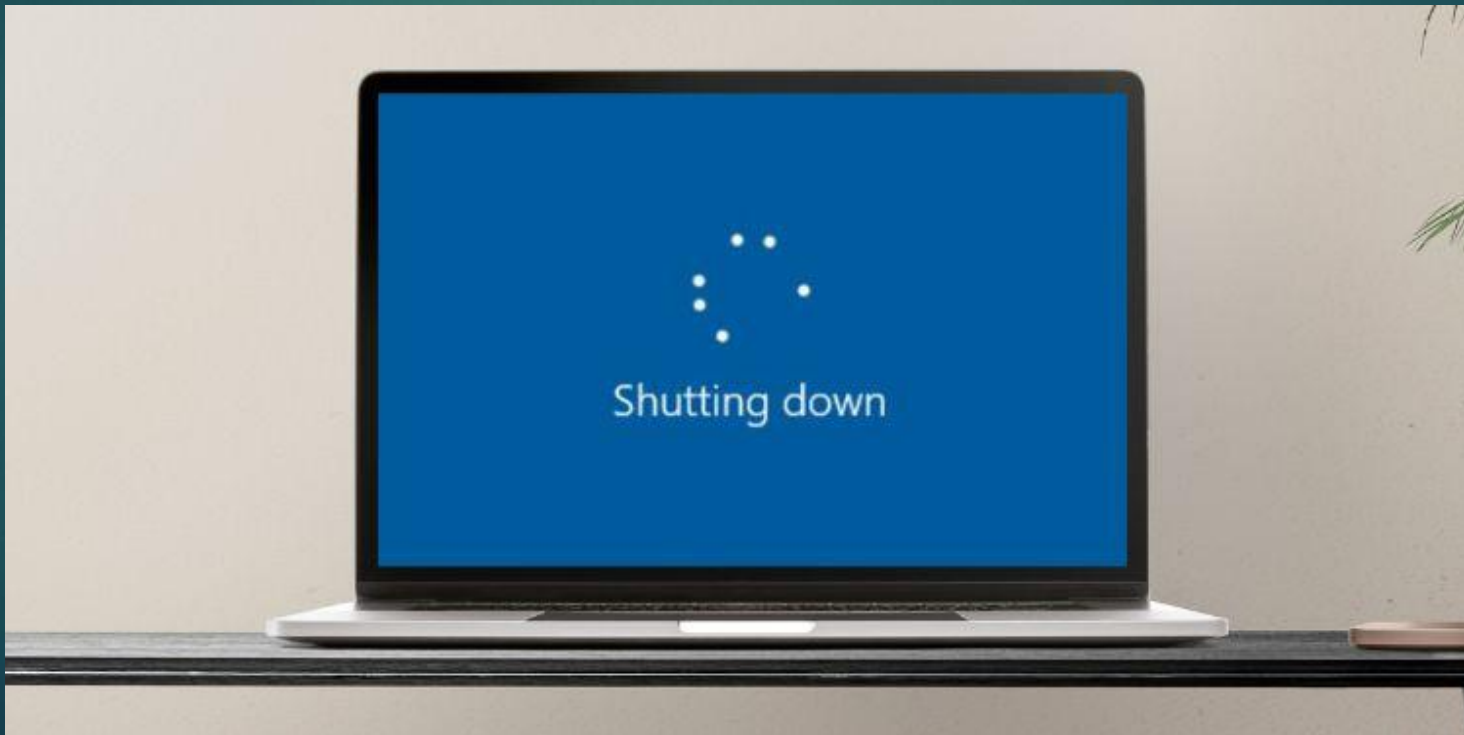
Windows Startup

- ▶ Occurs after the kernel is loaded
- ▶ System services and drivers are started
- ▶ User login screen appears
- ▶ Startup programs are launched after login



Windows Shutdown

- ▶ Closes all running applications
- ▶ Stops system services
- ▶ Saves settings and system state
- ▶ Safely powers off the hardware



Processes, Threads, and Services

- ▶ Process: a running instance of a program
- ▶ Thread: a smaller unit of execution inside a process
- ▶ Services: background processes that run without user interaction
- ▶ Managed by Task Manager and Service Manager

Memory Allocation and Handles

- ▶ Windows manages memory for each process
- ▶ Prevents programs from interfering with each other
- ▶ Handles are references to system resources
- ▶ Used for files, devices, and memory objects

The Windows Registry

- ▶ Central database for system and application settings
- ▶ Stores configuration information
- ▶ Organized into keys and values
- ▶ Critical for system operation

Registry Hives

- ▶ Major sections of the Windows Registry
- ▶ Examples: HKEY_LOCAL_MACHINE, HKEY_CURRENT_USER
- ▶ Contain hardware, software, and user settings
- ▶ Changes should be made carefully